

Transit Radio Cartographer

User Manual & Workflow Guide

A complete companion to the in-app tutorial.

Green Bank Observatory · Skynet Robotic Telescope Network

Contents

1. Overview & Conventions
2. Scan Workflow — measuring a single source
3. Survey Workflow — cleaning multi-sweep data
4. Pre-Image Workflow — reductions & gridding
5. Image Workflow — inspecting & saving the map
6. Flux Calibration Workflow — converting GCU to Janskies

This document mirrors the built-in tutorial in a single, printable form. The interactive tutorial inside the app remains the live reference; this manual is for reading start-to-finish, printing, or sharing as a PDF.

1 · Overview & Conventions

Transit Radio Cartographer builds radio maps from Green Bank 20m data: it calibrates, cleans, and grids individual sweeps into a final image. This manual walks through each stage of the workflow.

When you open another window while the in-app tutorial is showing, you may need to minimize and then maximize the Radio Cartographer window to bring the tutorial back into view.

The five workflows

- **Scan** — Open and measure a single source (.md1 → .scn).
- **Survey** — Open a multi-sweep survey, remove RFI, and accept sweeps (.md2 → .srv).
- **Pre-Image** — Smooth, baseline, align, and grid accepted sweeps into an image.
- **Image** — Adjust palettes, inspect flux, and save the final image.
- **Flux Calibration** — Build a measured-to-Jy calibration from known sources (.cal).

File types at a glance

Extension	What it is
.md1	Raw single scan from the telescope.
.md2	Raw multi-sweep survey from the telescope.
.scn	Processed scan (calibrated; may include a measured peak flux).
.srv	Processed survey (calibrated; per-sweep acceptance state preserved).
.img	Gridded survey image (binary raster produced from a survey).
.cal	Flux calibration table (measured-vs-known fit).

UI conventions used everywhere

- **Drag on a plot** selects a range (for cutting, declination filtering, peak fitting, or restoring removed samples).
- **Click a point** pins the readout () or sets an endpoint when a two-click tool (like Baseline Source or Remove RFI) is armed.
- **Click empty space** unpins or cancels a pending endpoint.
- **Hover** a point to see RA / Dec / Flux in the side panel.
- **Undo** reverts the most recent cut, removal, or restore.
- Tools are **sticky**: they stay armed after each operation so you can repeat without re-clicking the button.

2 · Scan Workflow

A *scan* is a single pass across one source, used mainly to measure that source's peak flux. The output (a `.scn` file) can later feed Flux Calibration.

1. Open the scan

1. Click `File → Open...` in the menu bar.
2. Pick a `.md1` file (raw scan) or a `.scn` file (already calibrated).

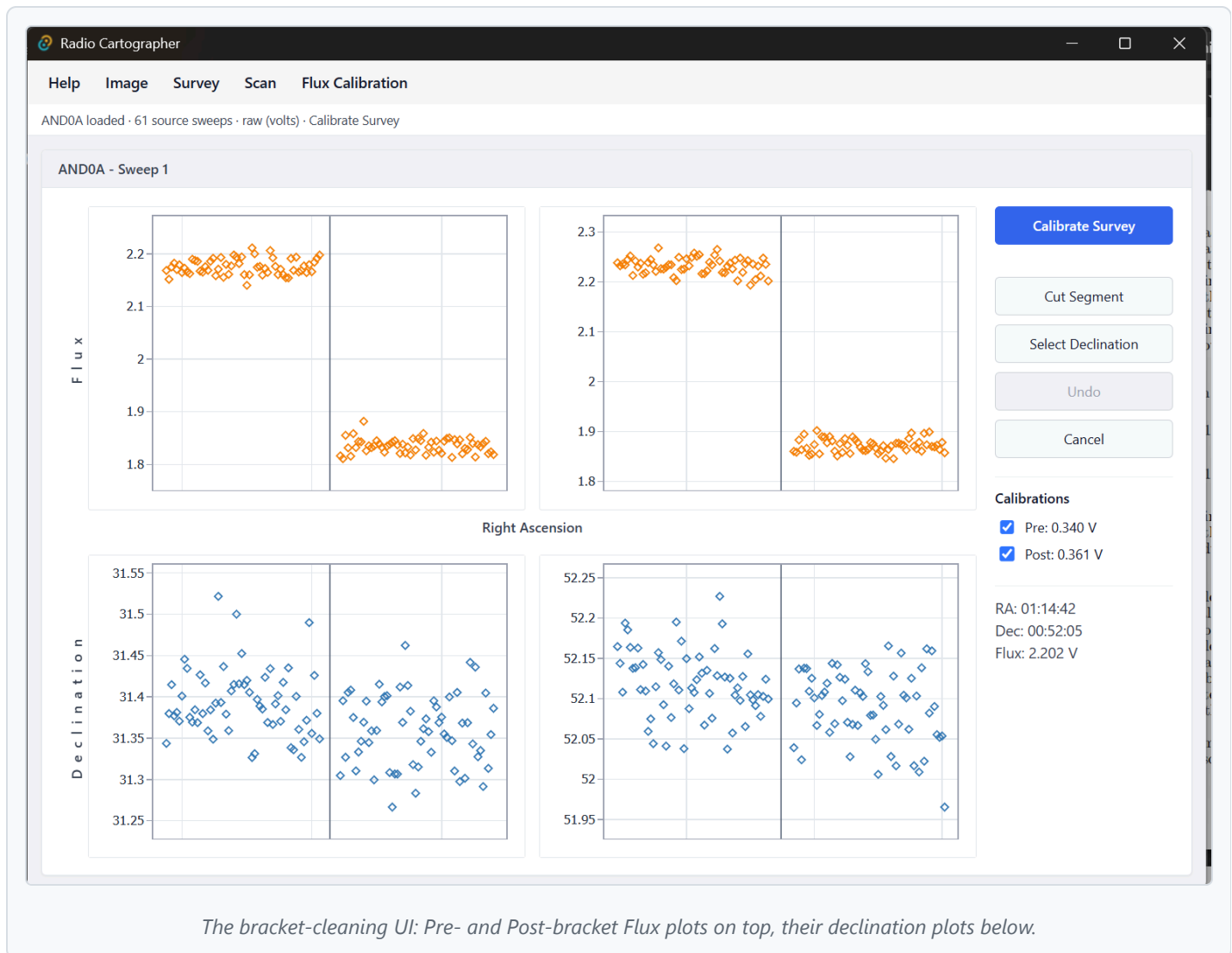
2. Inspect what loaded

Two stacked plots appear: **Flux vs RA** on top and **Declination vs RA** below. The cal-on, cal-off, and source samples are colored differently so you can tell where the calibration brackets begin and end.

3. Calibrate the scan

If you opened a raw `.md1`, click the **Calibrate Scan** button to enter the bracket-cleaning UI. Inside the calibration view you'll see four plots:

- Pre-bracket Flux (top left) and Post-bracket Flux (top right)
- Pre-bracket Dec (bottom left) and Post-bracket Dec (bottom right)



The tools here are:

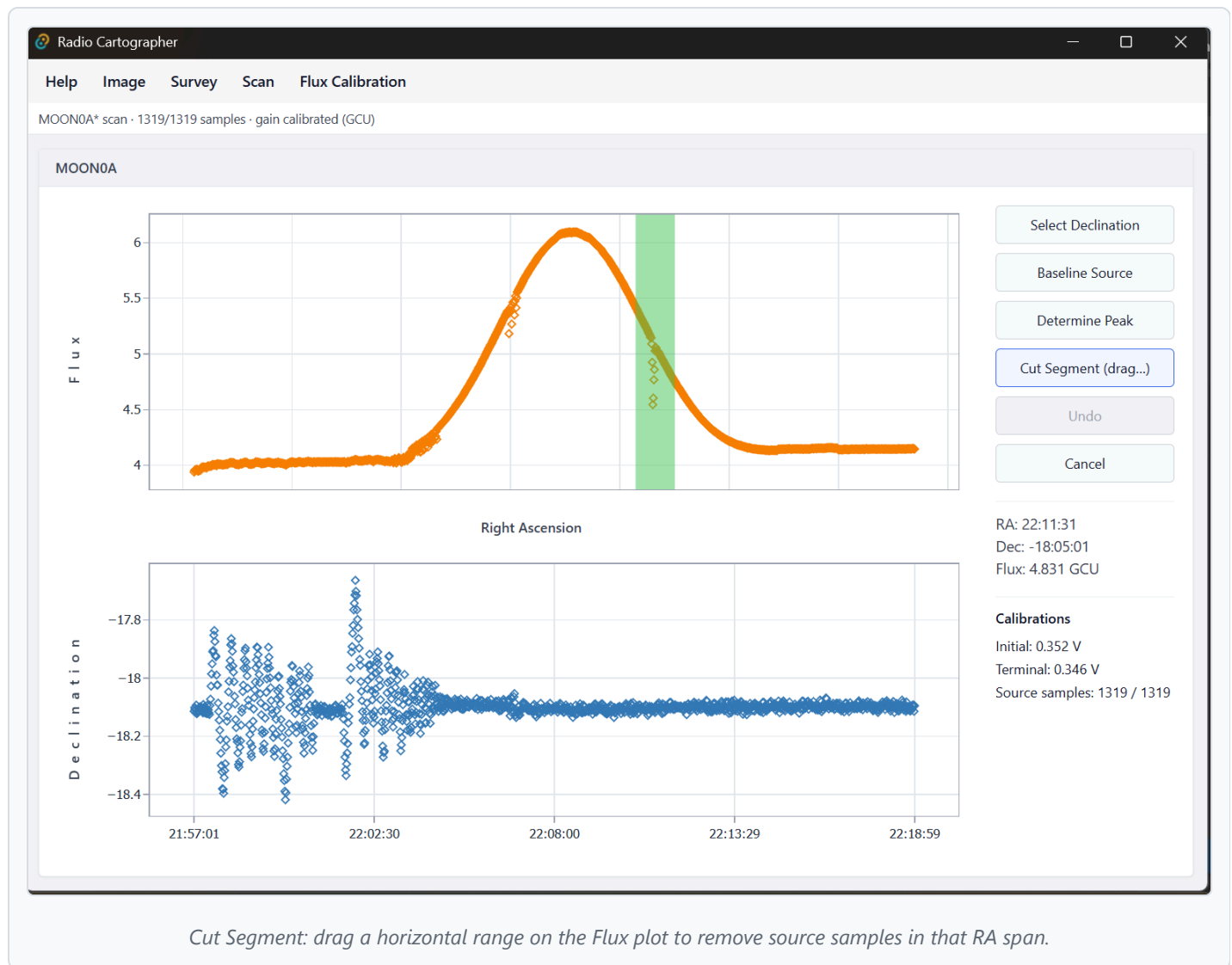
- **Cut Segment** — Click the button, then **drag horizontally** across an RA range on either Flux plot to delete samples inside that range. Tool stays armed for repeated cuts.
- **Select Declination** — Click the button, then **drag vertically** across a Dec range on either Dec plot to keep only cal samples within that band.
- **Pre / Post checkboxes** — Toggle a whole bracket off if it's corrupted beyond repair.
- **Undo** — Reverts the last cut or selection.
- **Calibrate Scan** — Applies the calibration and returns you to the main scan view.

4. Clean the source samples

Back in the main scan view, the source-reduction tools become available. Each button toggles a mode and stays armed until you click it again or switch tools.

4a. Cut Segment

Click the **Cut Segment** button, then **drag a horizontal range** on the Flux plot. Source samples in that RA range are removed.

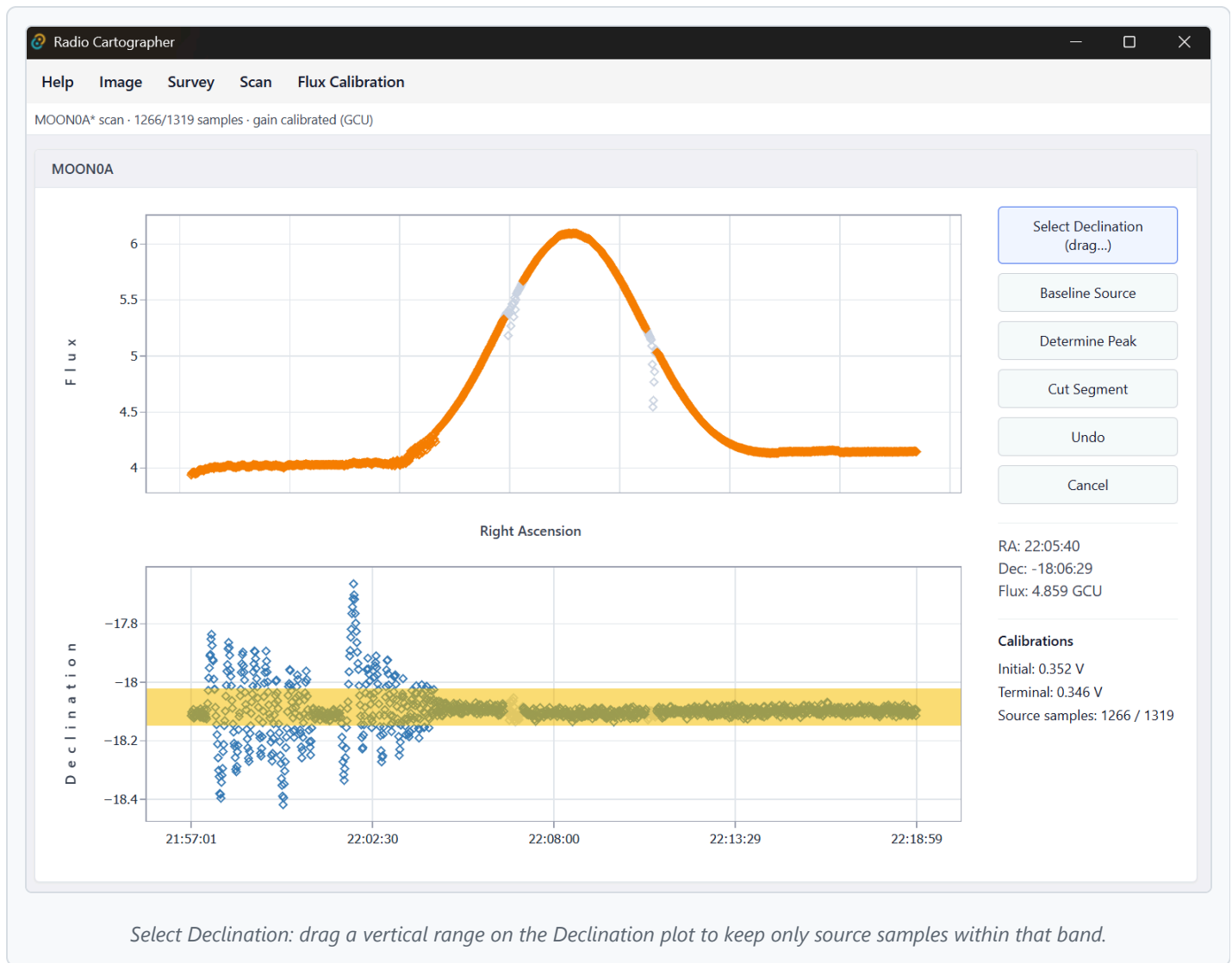


4b. Rescale

Cut samples stay on the plot as faded markers, so a strong RFI spike can keep dominating the flux scale even after you cut it. Click **Rescale** to fit the flux axis to the *kept* samples only — the source pops into view. The button stays lit while active and re-fits automatically after each further cut; **click it again** to return to the full scale.

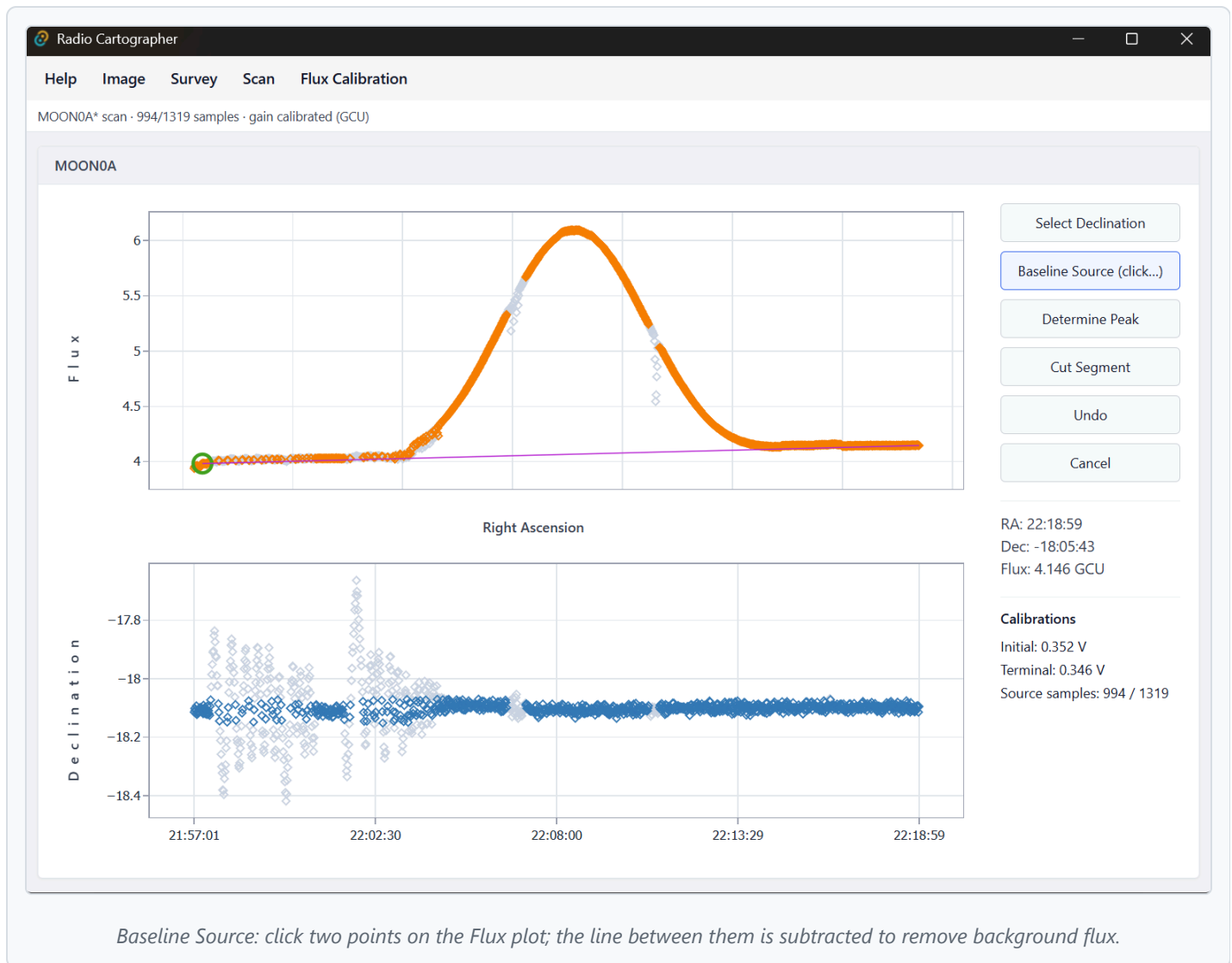
4c. Select Declination

Click **Select Declination**, then **drag a vertical range** on the Declination plot. Only source samples whose declination falls inside that band are kept.



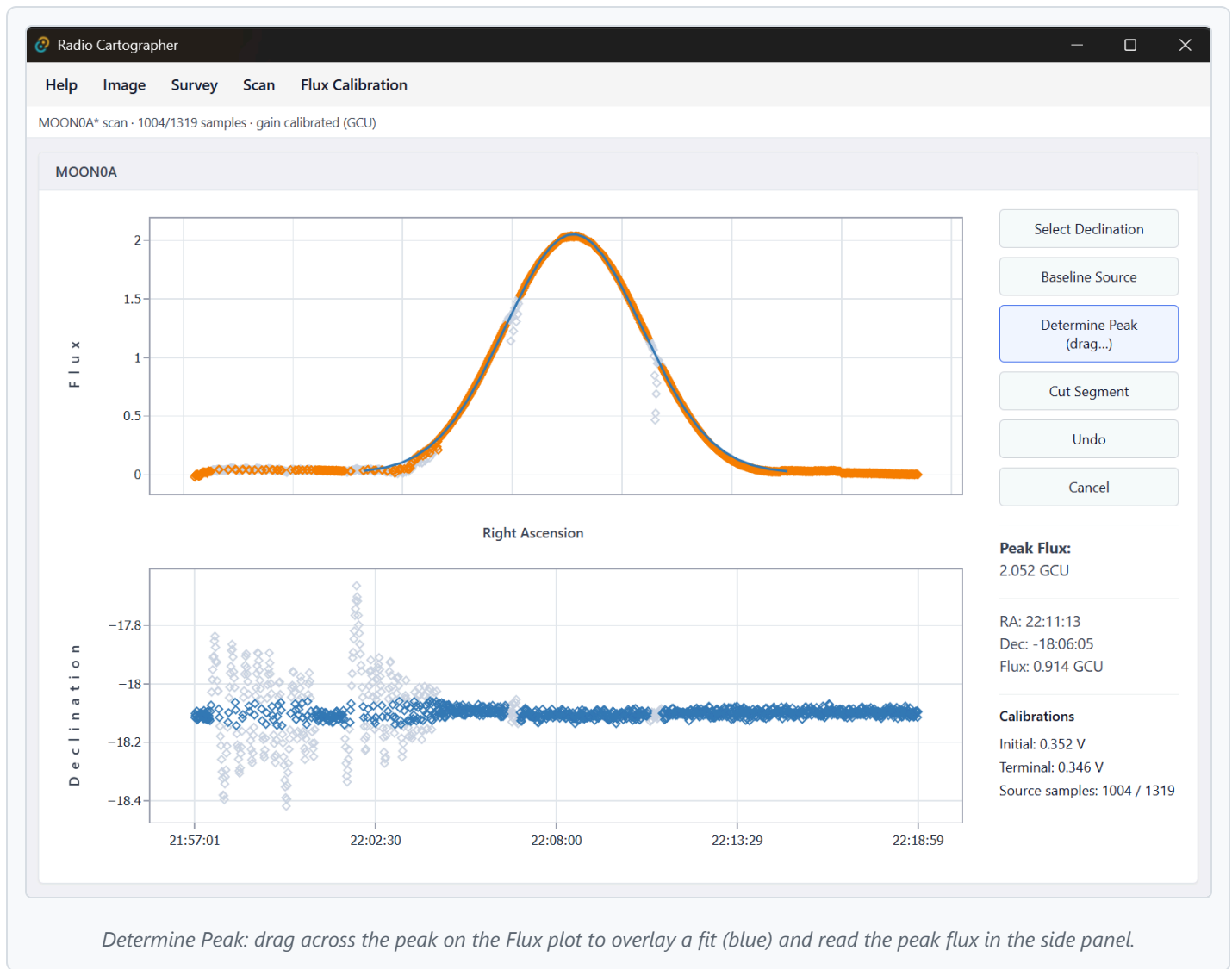
4d. Baseline Source

Click Baseline Source , then **click two points** on the Flux plot. A live rubber-band line follows your cursor between the first click and the second; the line between the two clicks is subtracted from the source, removing a background flux.



4e. Determine Peak

Click `Determine Peak`, then **drag a horizontal range** across the peak on the Flux plot. A Gaussian fit is overlaid in blue, and the peak flux appears in the side panel. Adjusting `Peak Fit Degree` underneath the scan menu allows for a 2nd, 3rd or 4th degree polynomial fit instead.



5. Hover and pin

Hover any point to read RA / Dec / Flux in the side panel. **Click** a point to pin the readout (a 📌 indicator appears). **Click empty space** to unpin.

6. Save your work

Click File → Save As... to write a .scn file. If you found a peak flux, it will be stored in the file header, so Flux Calibration can pick it up later without re-opening the scan.

3 · Survey Workflow

A *survey* is many sweeps over a patch of sky. The workflow is to calibrate, then walk through each sweep removing RFI, then move on to Pre-Image once every sweep is accepted.

1. Open the survey

1. Click **File** → **Open...** in the menu bar.
2. Pick a **.md2** file (raw) or a **.srv** file (in progress / partially accepted).

2. Calibrate the survey

If the file is uncalibrated, click **Calibrate Survey**. The bracket-cleaning UI is the *same* as Calibrate Scan: Cut Segment (horizontal drag), Select Declination (vertical drag), toggle Pre / Post, Undo, then click **Calibrate Survey** to apply.

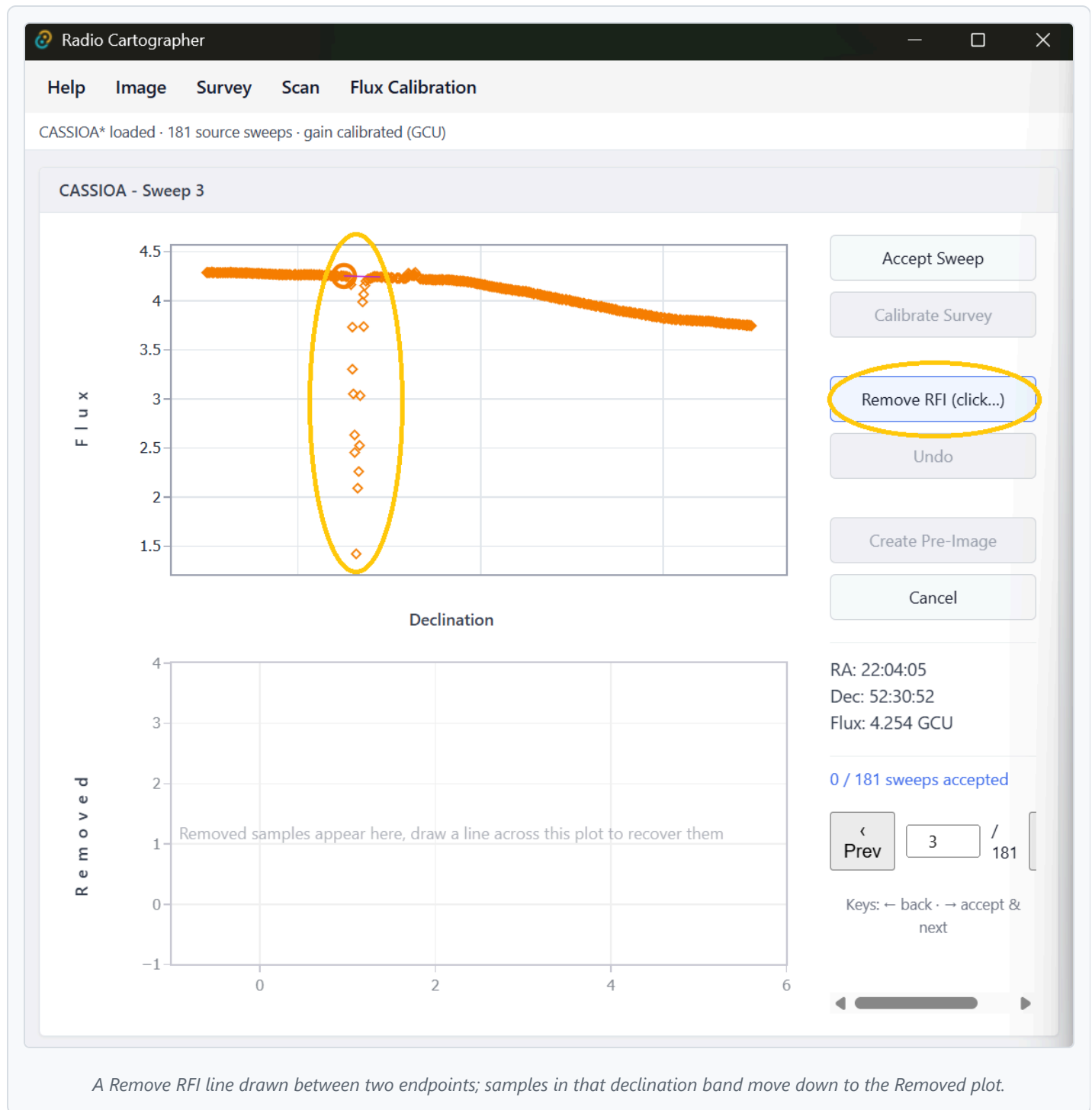


3. Sweep-by-sweep RFI removal

Each sweep has its own Flux plot (top) and Declination plot. At the bottom, the **Removed plot** shows samples you've taken out — empty until you remove something. You'll want to remove sharp dropouts and other

artifacts. For each sweep:

1. **Click** Remove RFI . The button label changes to Remove RFI (click...) to confirm it's armed.
2. **Click one endpoint** on the Flux plot. A pending marker appears.
3. **Click a second endpoint**. A line is drawn between the two points and samples inside that declination band are moved to the Removed plot. The tool stays armed so you can keep removing.



Hover any point to read its RA, Dec, and flux in the side readout, and **click** a point to pin those values (a  appears; click empty space to release). The readout keeps updating as you click, even while **Remove RFI** is armed, so you can check exactly which samples a band will take out before committing.

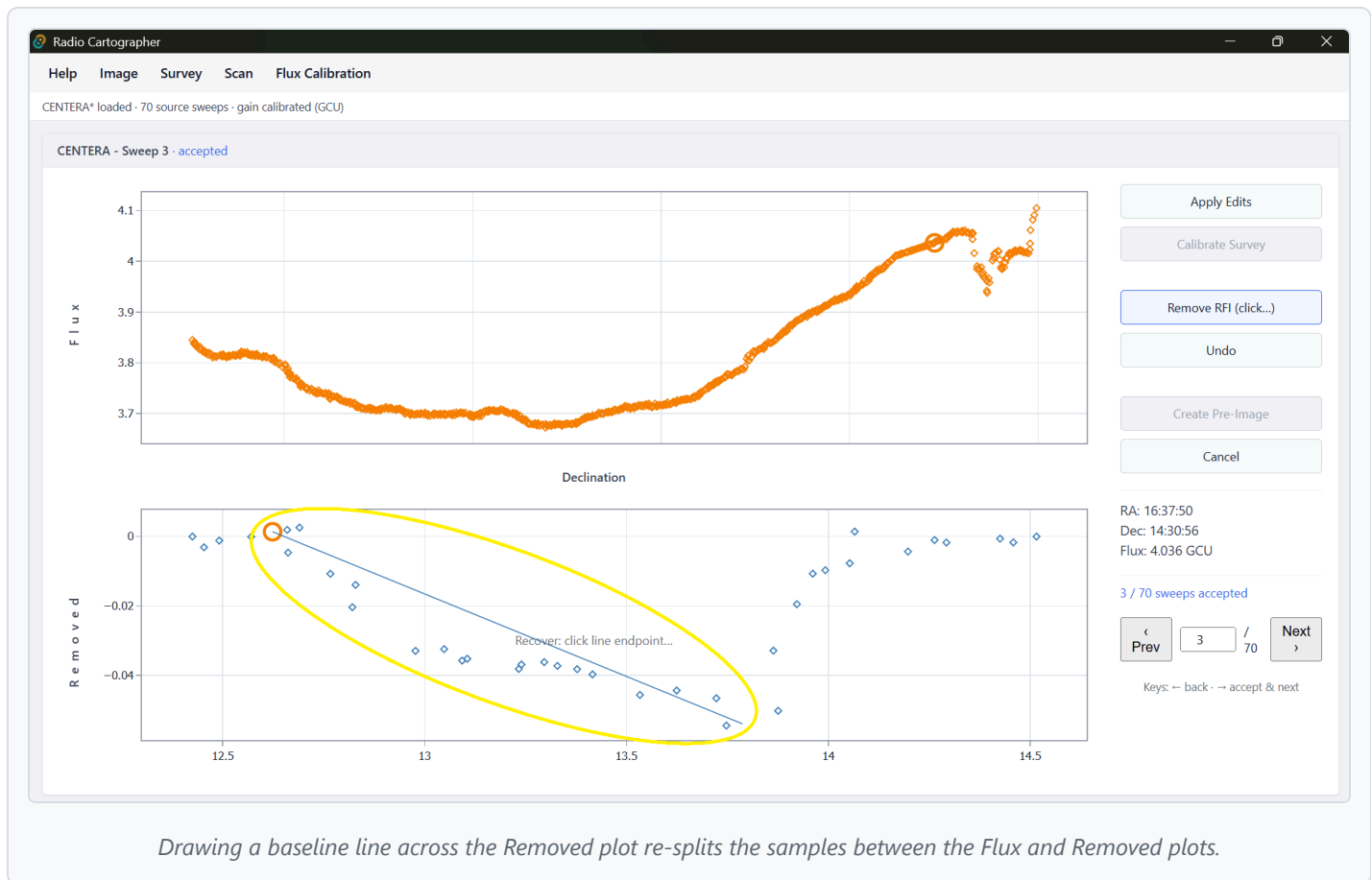
3a. Undo a removal by re-drawing the baseline

The Flux plot and the Removed plot always add up to the original data — removing RFI just moves structure from the top plot down to the bottom one. Restoring works the same way in reverse, and it isn't a "put the samples back" button: you **draw a new baseline line across the Removed plot** and the two plots re-split along it.

With **Remove RFI** still armed (and at least one sample already removed), use the same two-click gesture on the bottom panel: click one endpoint, then a second. The line is free, not snapped to a sample. For every removed sample whose declination falls inside the line's span, the removed residual is laid onto the line you drew and the part above the line flows back into the Flux plot:

- Draw the line **along zero** to send the full residual back to the flux — this is the case that reproduces the original samples.
- Draw it **through the removed points** to leave them as they are (nothing moves).
- Draw it **anywhere between** to split the difference — the flux keeps what sits above your line, the Removed plot keeps the rest.

Undo also reverts the most recent removal or restore.



3b. Accept the sweep

When the sweep looks clean, click **Accept Sweep**. The sweep is marked accepted (a tag appears next to the file name and sweep number) and pending edits are committed. When you click **Accept Sweep** it will automatically advance to the next unaccepted sweep.

Accepting isn't final — you can navigate back to an accepted sweep and its removed samples are still shown on the Removed plot. Remove more RFI or draw a baseline to restore points just as before; the accept button now reads **Apply Edits** and re-commits your changes without un-accepting the sweep. (This history is per session: it lasts while the survey is open, but reopening a saved `.srv` starts each sweep from its committed flux with an empty Removed plot.)

4. Navigate sweeps

Use the **Prev / Next** buttons or **type a sweep number** in the sweep input field — these just move between sweeps and do *not* accept anything. This allows you to navigate between sweeps without committing changes. The progress indicator `N / M sweeps accepted` updates live.

Keyboard shortcuts (active whenever the sweep-number field isn't focused): `←` moves to the previous sweep, and `→` *accepts* the current sweep and advances once the survey is calibrated (before calibration it just moves forward).

Accepting a sweep jumps you to the next *unaccepted* one and wraps around — so if you accept out of order, accepting the last sweep loops back to the first sweep you still need to finish. That wrap is how you track down any stragglers before the image can be built.

5. Save partway through (optional)

Click `File → Save As...` to write a `.srv`. You can come back later, reopen the `.srv`, and resume on the next unaccepted sweep. Note that if you save a `.srv`, upon re-open you will no longer be able to restore removed points from previously accepted sweeps. You will still be able to return to the accepted sweeps and remove more points.

6. Move on to imaging

Once *all* sweeps are accepted, the **Create Pre-Image** button enables. Click it to continue to the Pre-Image stage.

4 · Pre-Image Workflow

Pre-Image takes all your accepted sweeps and runs three reductions on them — smooth, baseline, align — before gridding into a final image. The three reductions **must be run in order**, and **Make Image** only enables once all three are done.

A preview at the default pixel size (0.06°) is generated automatically when you arrive, so you can see how the reductions change the result. **Drag a box** on the preview to zoom into a region (any shape — the axes fit exactly what you draw) and **double-click** to reset to the full view; this is handy for checking whether a reduction cleaned up a particular patch of sky.

1. Smooth Sweeps

Click **Smooth Sweeps** . The smoothing window is fixed at 5 samples; the preview redraws when it's done. No dialog — one click and you're done.

2. Baseline Sweeps

1. **Click** **Baseline Sweeps** .
2. A dialog asks for **Baseline Length (Degrees)**. Default is 5° .
3. **Type** a value (or accept the default), then click **OK** .

3. Align Sweeps

1. **Click** **Align Sweeps** .
2. A dialog asks for **Maximum Declination Shift (Degrees)**. Default is 0.5° .
3. **Type** a value, click **OK** .

4. Make Image

1. Confirm all three reduction buttons show their "done" state — only then does **Make Image** enable.
2. **Click** **Make Image** .
3. A dialog asks for **Pixel size (degrees)** — the on-sky size of each output pixel. The default 0.06° is 1/20 of the 40 ft beam (1.2°). **Smaller is finer**: a smaller value gives more, finer pixels; a larger value coarsens the grid into fewer, blockier pixels. The default tracks your previous run.
4. **Type** a value, click **OK** . The final gridded image is built and you'll switch to the Image view.

5. Back to Sweeps (if needed)

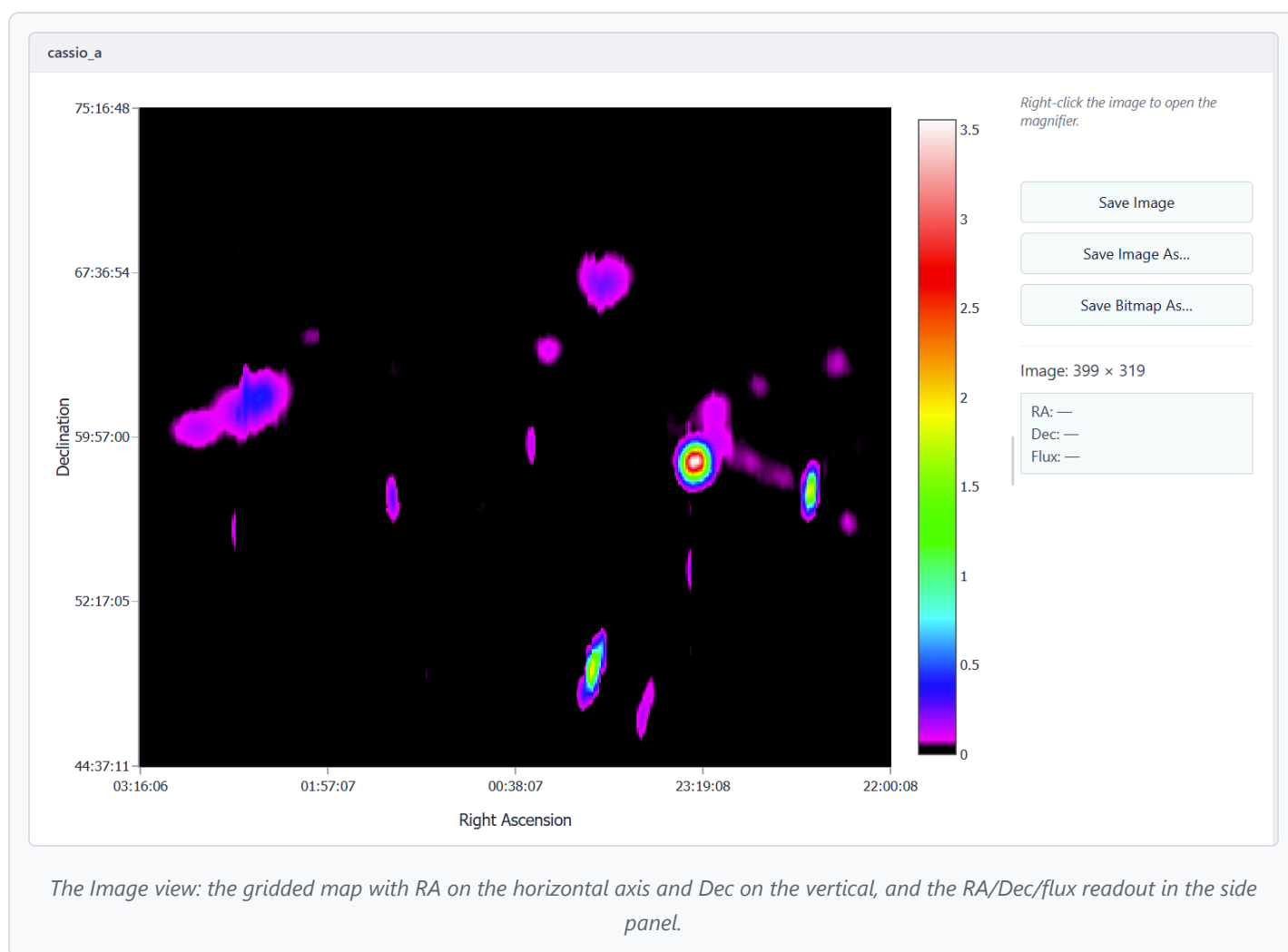
If the preview reveals some RFI slipped through, click **Back to Sweeps**. Your sweep acceptances are preserved, so you can fix it, hit **Apply Edits** , and come back to redo the reductions.

5 • Image Workflow

The Image view is where you inspect the final gridded radio map — pan and zoom into it, pin and magnify pixels to read off values, tune the palette for publication-quality figures, and save or compose outputs.

1. Inspecting the image

After **Make Image** runs, the gridded map appears as a 2D color image, with Right Ascension on the horizontal axis (increasing to the left, as on the sky) and Declination on the vertical. **Hover** any pixel to see its RA, Dec, and flux in the readout panel on the right; that panel also shows the image dimensions. The readout keeps its last value when the cursor leaves the image.



1a. Zoom and pan

Drag a box anywhere on the image to zoom the axes to that region — any shape works, the view fits exactly what you draw. **Double-click** to reset to the full map. The default view preserves the true sky shape; because zooming is free, a zoomed-in region may look stretched, which is expected.

1b. Pin a pixel

Left-click a pixel to **pin** it. A target ring marks the spot and the readout locks to that pixel's RA, Dec, and flux (tagged *pinned*) so the numbers stay put while you move the mouse or zoom elsewhere. Click **Unpin** — directly under the readout — to release it. If you zoom to a region that doesn't contain the pinned pixel, the readout still shows its values even though the ring is off-screen.

1c. Magnifier

Right-click a pixel to open the **magnifier** — a zoomed inset of the region around that spot, shown in the side panel. Use the **arrow keys** to nudge the magnified area (hold **Shift** for ×5 steps), and **Close Magnifier** to dismiss it. Change how many cells it covers with **Image → Change Magnifier Size...**. Right-clicking only opens the magnifier — it never disturbs your current zoom.

While the magnifier is open, three buttons below the RA/Dec/Flux readout let you total the flux in and around the magnified box:

- **Sum Flux (Box)** — adds up the flux of every covered pixel inside the magnifier box, and shows how many pixels went into the sum.
- **Average Flux (Box)** — the mean flux over those same covered pixels (no-coverage cells are ignored, so blank sky in the box doesn't pull the average down).
- **Sum Flux (Outside Box)** — the total flux of the whole map *excluding* the magnifier box, useful for separating a source from its surroundings.

Each button toggles its value on or off, and the numbers update **live** as you move the box with the arrow keys or right-click a new spot — so you can watch a total change as you slide the box across a source. Values use the same flux unit as the pixel readout.

1d. Display shape

Image → Image Display ▶ sets the aspect ratio of the default (un-zoomed) view:

- **Declination Corrected** (default) — applies a $\cos(\text{declination})$ correction at the image center so the displayed shape matches the true sky.
- **No Declination Correction** — RA-seconds → degrees only (1/240); accurate at the celestial equator.
- **Snap to Square** — each pixel cell renders square on screen; handy for inspecting very thin or very wide surveys.
- **Stretch to Fill** — no aspect lock; the map stretches to fill the workspace area.

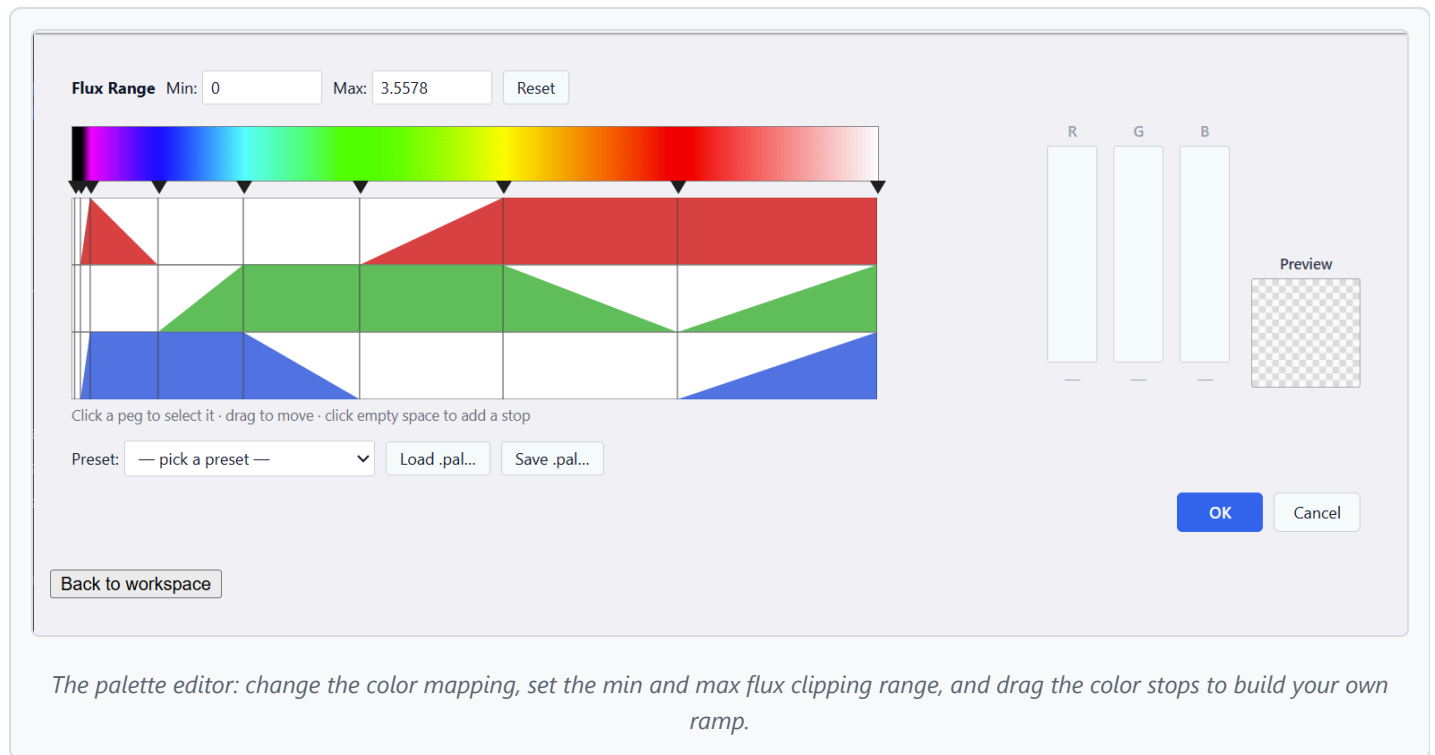
2. Find the true peak

The brightest-colored pixel isn't always the maximum flux pixel — if the palette range clips, several pixels may share the top color. Sweep the cursor over the suspected peak region and watch the readout, or open the

magnifier on it, to identify the true maximum; then **pin** it so its value stays on screen while you set the palette range or compare candidates.

3. Tune the palette

Open the palette controls with **Image → Show Palette...** (enabled once an image is built or loaded). From there you can change the color mapping, the min and max flux clipping range, and — for composites — the channel colors.



3a. Adjust max flux

Type the maximum flux value. Anything above it clips to the top color of the palette. Set it to your true peak (from step 2) to maximize dynamic range.

3b. Adjust min flux

Type the minimum flux value. Anything below it clips to the bottom color.

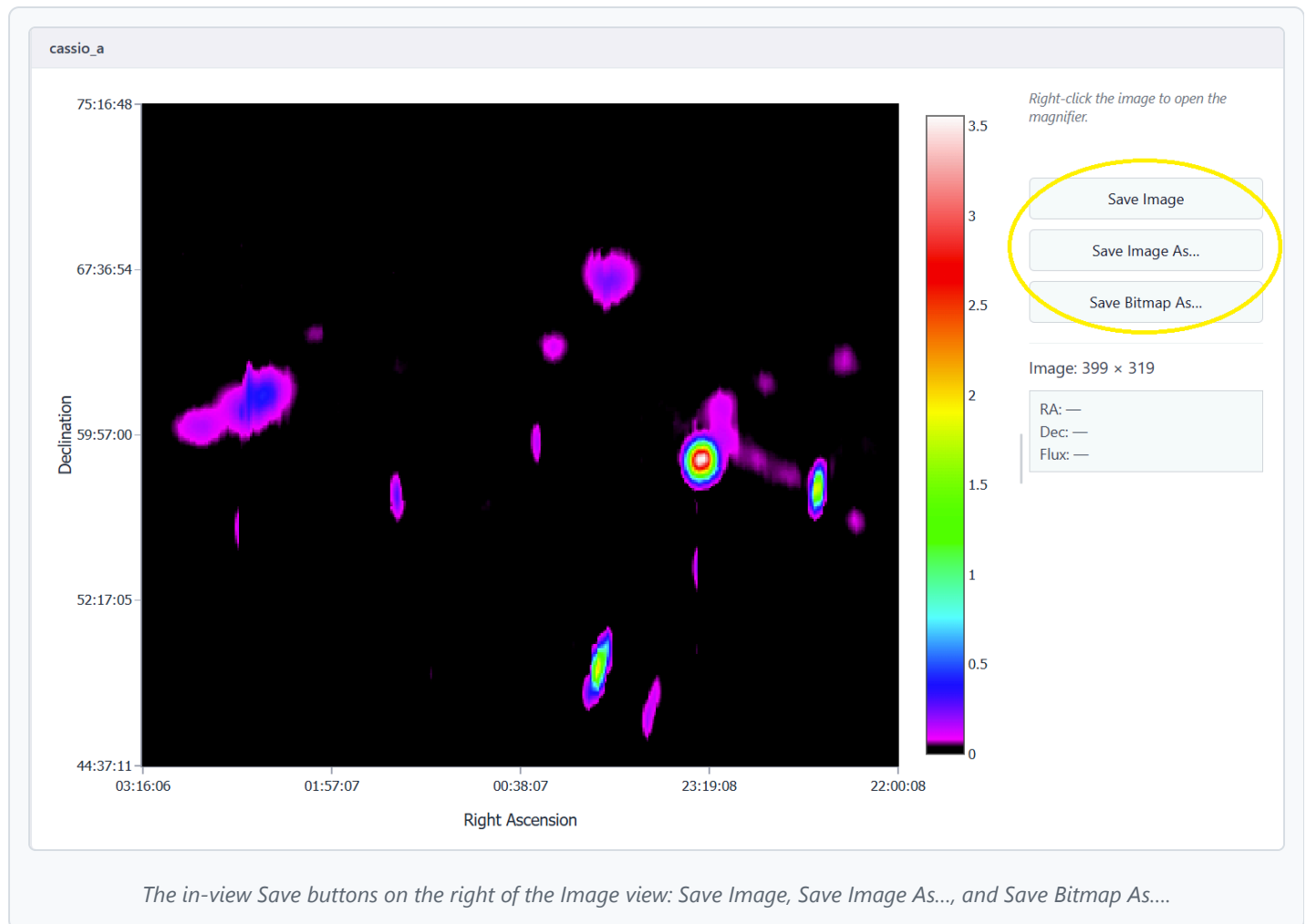
3c. Pick a palette

There are default presets, or you can load in a .pal file. You can make your own by dragging the color stops. Stops stay ordered and spaced automatically, so you can add many of them or drag them to the edges without breaking the ramp. If you want to save the palette for later, click **Save Palette...** to write a .pal file.

4. Save the image

The scalar Image view has in-view buttons on the right — **Save Image** , **Save Image As...** , and **Save Bitmap As...** — so you can save without opening a menu. The same actions are also on the **Image** menu:

- **Save Image** — Saves the app-native image format back to the current path.
- **Save Image As...** — Saves to a new path.
- **Save Bitmap As...** — Exports a **.png** / **.bmp** for use in papers / slides.



5. Compose multiple images

- **Image → Append Image...** — Stitch scalar maps side by side; you can pick several at once. Where footprints overlap, each cell takes the **maximum** flux of the covering maps.
- **Image → Superimpose Image...** — Overlay scalar maps in the same field; you can pick several at once. Where footprints overlap, each cell takes the **average** of the covering maps. Two maps can be blended with a custom weight; three or more are always weighted **evenly** — to adjust individual weights, superimpose one map at a time.
- **Image → Make Bi-Color Image...** — Assigns two scalar maps to two color channels for a two-band composite.

- **Image → Make Tri-Color Image...** — Three-band RGB composite. If you already have a bi-color image, this can extend it by adding the unused channel.

A composite (bi/tri-color) opens in the same view. It supports the same **drag-to-zoom** and **right-click magnifier**, and hovering shows an RA/Dec readout. Use **Export as PNG...** to write the composite out as an image for papers or slides, and **Back to Image** to return to the scalar map it was built from.

6 · Flux Calibration Workflow

Flux Calibration converts calibrated machine output (GCU) into Janskies (Jy) by fitting a slope through one or more known calibrators. The output is a `.cal` file that the rest of the app can use.

Applying an existing calibration to your workspace

If you already have a `.cal` file that fits your data (same telescope, epoch, and frequency band), you don't need to build a new one — just load it and the app converts whatever is currently in your workspace from GCU into Janskies for you.

1. Make sure the workspace you want to convert is already **calibrated** (in GCU). Loading a `.cal` converts GCU to Janskies, it does not perform gain calibration.
2. **Click** `Flux Calibration → Select Calibration...` and pick your `.cal` file.
3. The app reads the fitted slope (Jy/GCU) from the file and **automatically applies it** to whatever is open — the survey, the current scan, and any generated image — so their values are now in Janskies. Already-flux-calibrated workspaces are skipped, so re-loading won't double-convert.

This is the common path once a good calibration exists: build the `.cal` once (steps below), then just load it for every other data set.

1. What you need first

Each calibrator must be a scan that already has a measured peak flux. If a scan's peak hasn't been measured yet, open it in the Scan view and run **Determine Peak** first (see the [Scan](#) section).

2. Open Flux Calibration

- To start a fresh table: `Flux Calibration → New Calibration...`
- To edit an existing one: `Flux Calibration → Select Calibration...` and pick a `.cal` file.

3. Name your calibration

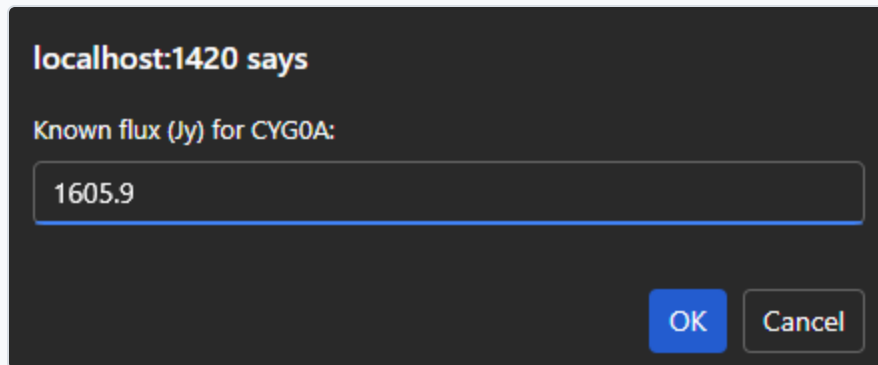
At the top of the view there's a caption field — **type** a descriptive name like `"cyga_2026"`. This is stored in the `.cal` file so you can tell calibrations apart later.

4. Add calibration sources

4a. From a saved .scn file

1. **Click** `Add Source from File...`
2. Pick a `.scn` file. (If its header has no peak flux, the app will tell you to run Determine Peak first.)

3. A prompt appears: **Type** the known flux of that source in Janskies, then **OK**. Taurus A (Crab), Virgo A, and Cygnus A are our best calibrators, but any source with a known flux is valid. Those three sources will automatically populate the prompt with their known fluxes. The 40 foot receiver operates at 1350 MHz to 1430 MHz.



localhost:1420 says

Known flux (Jy) for CYG0A:

1605.9

OK Cancel

Adding a source: after picking a `.scn` file, enter the source's known flux in Janskies.

4b. From the currently open scan

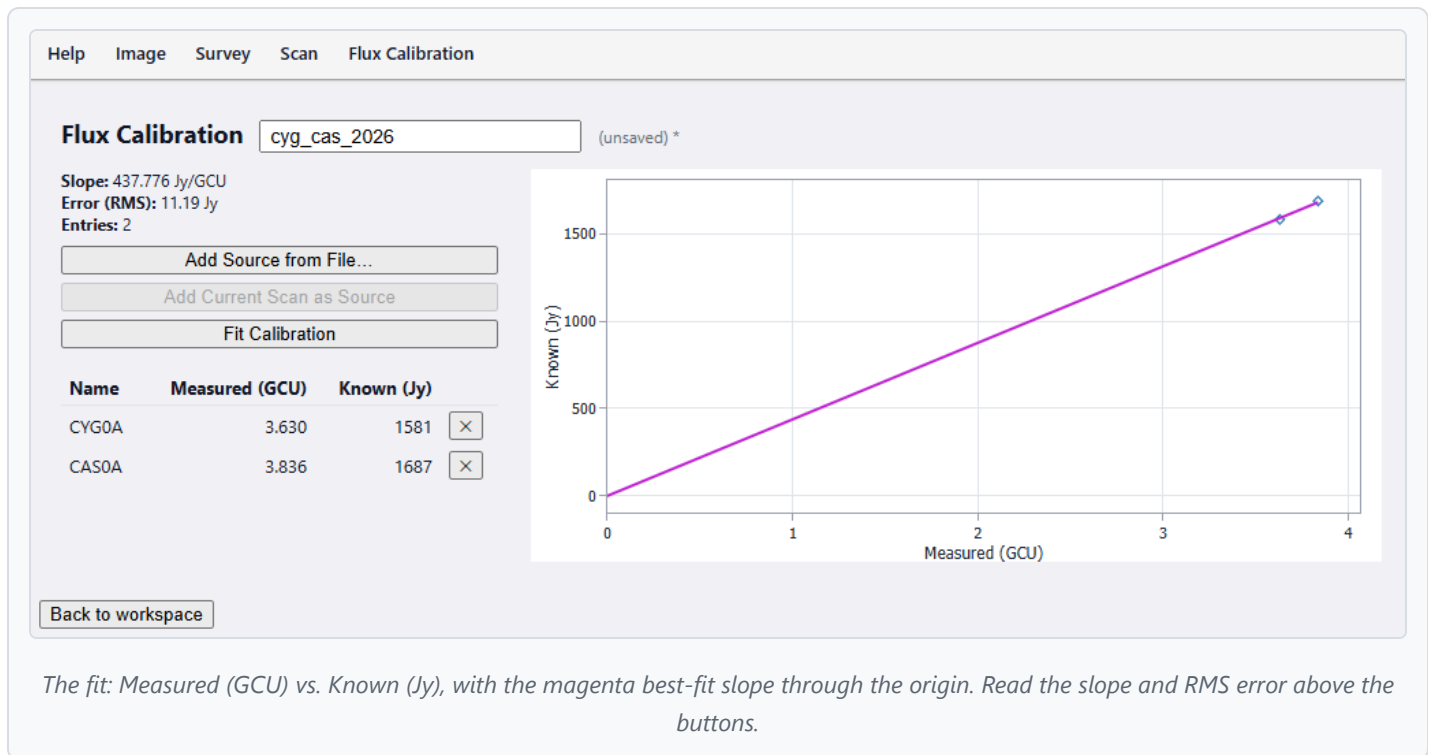
If you already have a scan open in the Scan view with a measured peak, click **Add Current Scan as Source**. A prompt asks for the known flux just like above.

4c. Remove an entry

Each row in the calibrator table has a **×** button. **Click** it to drop that calibrator from the fit.

5. Fit the calibration

1. **Click** **Fit Calibration**.
2. The scatter plot shows **Measured (GCU)** on the X-axis vs. **Known (Jy)** on the Y-axis. A magenta line is the best-fit slope through (0,0). You do not need multiple calibrators to fit a slope, but more points give a better estimate of the slope and its error.
3. Read the slope (Jy/GCU) and RMS error from the panel above the buttons. A large RMS relative to the slope is a hint that one of your calibrators is bad — drop it and refit.



6. Save the .cal

Click **File** → **Save As...** to write the calibration to disk.